Introduction

SHAP (SHapley Additive exPlanations; Lundberg & Lee, 2017) attributes each prediction to input features via the Shapley value from cooperative game theory. It is the only attribution method satisfying the combined axioms of local accuracy, consistency, and missingness. SHAP has become the standard for black-box model interpretability.

Prerequisites

Variable importance; cooperative game theory basics helpful.

Theory

For prediction $f(x)$, each feature j gets a SHAP value ϕ_j satisfying

$$f(x) = \mathbb{E}[f(X)] + \sum_j \phi_j.$$

Shapley values average the marginal contribution of feature j across all possible subsets of features, producing a fair attribution. Exact computation is exponential; Tree SHAP (Lundberg, 2020) computes it in polynomial time for tree ensembles.

Assumptions

Features have sensible "absence" interpretations (TreeSHAP integrates over feature distributions); additive decomposition meaningfully captures model behaviour.

R Implementation

```r
library(xgboost); library(shapviz)

set.seed(2026)
n <- 500
X <- matrix(rnorm(n * 4), n, 4); colnames(X) <- paste0("x", 1:4)
y <- as.numeric(X[, 1] > 0) + 0.5 * as.numeric(X[, 2] > 0) +
     rnorm(n, 0, 0.5)

dtrain <- xgb.DMatrix(X, label = y)
model <- xgb.train(list(objective = "reg:squarederror",
                        max_depth = 3, eta = 0.1),
                   data = dtrain, nrounds = 200, verbose = 0)

sv <- shapviz(model, X_pred = X, X = X)

# Summary (importance ranking)
sv_importance(sv)

# Per-observation waterfall (first prediction)
sv_waterfall(sv, row_id = 1)
```

Output & Results

An importance plot (mean |SHAP| per feature) and a waterfall plot showing how each feature pushes the specific prediction above or below the baseline.

Interpretation

"SHAP ranked x_1 as the dominant driver (mean |SHAP| = 0.36); the waterfall showed observation 1's prediction was pushed up by 0.42 from baseline primarily due to $x_1 = 1.8$."

Practical Tips

- Use **Tree SHAP** (fast, exact) for tree models; fallback to **Kernel SHAP** for arbitrary models (approximate, slower).
- SHAP dependence plots (SHAP value vs feature value) reveal non-linear effects and interactions.
- SHAP values sum to $f(x) - \mathbb{E}[f(X)]$, so per-observation explanations are additive.
- For deep learning, use `DeepSHAP` or integrated gradients; TreeSHAP does not apply.
- Global importance = mean |SHAP| across observations; usually close to permutation importance on tree ensembles.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Cross-validation, nested CV, bootstrap .632+
- Regularisation: ridge, lasso, elastic net
- Trees, random forests, gradient boosting
- Interpretability and SHAP
- Tabular neural networks with torch
- Imaging and sequence models intro
- tidymodels pipelines
- FDR, knockoffs, replication crisis